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Sub-Quadratic Sequence Modeling with Structured Linear State Space Projections

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ABSTRACT

The quadratic computational complexity of standard multi-head self-attention restricts transformer context windows on long research documents. We propose a sub-quadratic sequence modeling architecture based on structured linear state space projections and diagonalized convolution kernels. Our model processes sequences up to 64,000 tokens with linear time and memory complexity while maintaining expressive associative recall. Across synthetic associative recall and language modeling benchmarks, our method matches vanilla Transformer-XL perplexity while reducing memory consumption by 72%. However, the model exhibits degraded performance on dense cross-document token retrieval requiring precise non-local associative lookups.

Keywords— Machine Learning, Transformers, Sequence Modeling, State Space Models

INTRODUCTION

Self-attention's O(N^2) complexity imposes severe bottlenecks when processing entire academic papers or book-length corpora. Linear attention approximations often sacrifice expressivity. We present a principled state-space formulation achieving exact linear scaling.

METHODOLOGY

We discretize continuous-time state-space representations using HiPPO matrix initializations. By utilizing associative scan algorithms on modern GPU hardware, sequence processing runs in O(N) time without explicit attention matrices.

RESULTS

On long-context scientific text modeling (16k to 64k tokens), our model achieved a 4.1x inference speedup while utilizing 72% less peak VRAM compared to FlashAttention baselines.

LIMITATIONS

The continuous projection struggles with discrete token-copying tasks where needle-in-a-haystack lookups require exact point-to-point token attention rather than smoothed latent representations.

REFERENCES

[1] Vaswani, A., et al. (2017). "Attention Is All You Need". Advances in Neural Information Processing Systems, 30.

[2] Kipf, T. N., & Welling, M. (2016). "Semi-Supervised Classification with Graph Convolutional Networks". arXiv:1609.02907.

[3] Lewis, P., et al. (2020). "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks". NeurIPS.